

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS -DECEMBER 2025
DATA ARCHITECTURE [DS_3151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) "BioMed Corp" processes complex patient medical history forms containing both structured data and long, doctor-written narrative sections. (5)
- A) Critically evaluate why Textual Disambiguation (Textual ETL) is absolutely necessary to process the doctor's narrative text, focusing on the core challenge of context derivation in non-repetitive text, rather than relying on traditional ETL methods used for structured data.
- Course Outcome: "CO1" / Cognitive Level: "Evaluate"
- B) Analyze the use of Associative Word Processing and Proximity Analysis in this medical context. Explain how these techniques help BioMed infer context by examining related terms appearing close to each other within the narrative text. (3)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- C) Define the primary differences between Structured Data and Unstructured Data based on their predictability and reliance on a DBMS infrastructure for management and accessing individual data units. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) The Chief Architect at "InnovateSoft" is implementing the Zachman Framework to improve compliance. Developers currently struggle to connect architectural goals to their daily tasks, perceiving the architecture as "ambiguous". (5)
- A) Critically evaluate two key principles of the Zachman Framework: Unambiguous Artifact Placement and Completeness of Architecture. Explain how rigorous adherence to these principles could directly mitigate the perception of the architecture being "ambiguous."
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Analyze the pivotal role of the Information Architect in addressing this ambiguity. Explain why the architect must possess the ability to grasp abstract concepts (like the overall blueprint) while also descending to the necessary level of detail to articulate a problem's impact on the entire system. (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Apply the foundation principles of Data Governance by defining three Key Data Policy Principles that InnovateSoft should mandate to ensure architected development produces repeatable results. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) An agency needs to formalize its logical models using a Logical Modeling Tool and track complex employee dependencies and history. (5)
- A) Evaluate the logical modeling choice of treating History as an Associative Entity. Explain why this approach is preferred for keeping the core non-time-dependent entity pure and intact in the logical model.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) During database design for a human resources system, the data architect discovers that the Employee entity has both a recursive relationship and a circular reference involving the Department and Project entities. (3)
- Analyze how denormalisation can be used to resolve these two issues. Explain why recursive relationships often require additional foreign keys, while circular references may need the creation of a new fourth normal form (4NF) entity to eliminate dependency loops.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) In a customer management system, the company wants to maintain a record of each time a customer's contact information changes without deleting or overwriting previous data. (2)
- Apply the concept of Simple History to this case. Explain how the system should mark the previous record when a new version is added, and describe what attribute can be used to identify the most recent record.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) The agency DBA must convert the logical model to a physical model for an OLTP database, potentially compromising normalization for performance. (5)
- A) Evaluate the primary function of the Logical Modeling Tool. Explain the role of the Data Administrator (DA) in using this tool to decompose data elements and work activities to the elementary level, ensuring the resulting model is impartial to usage.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) During the conversion of a Logical Data Model (LDM) to a Physical Data Model (PDM), the DBA considers adding a few Derived Attributes (like total sales or average balance) to improve query performance. However, this introduces data redundancy. (3)
- Analyze the trade-off involved in this denormalization decision. Explain why adding such redundancy is acceptable only when it is supported by access path analysis, and how improper justification could affect data consistency and maintenance overhead.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) A company is beginning a data modernization project. The architects first create a Conceptual Business Model (CDM) to represent business entities like *Customer*, *Order*, and *Product*, and later convert it into a Physical Model (PDM) for implementation on a relational database. (2)
- Apply your understanding to explain the key difference between the CDM and the PDM in this situation. Which model is strategic and abstract, and which is technology-dependent and detailed? Provide one example of how an entity or relationship might change between the two.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) The DBA is finalizing the Physical Data Model (PDM) for a high-volume transaction system. To improve performance, the team proposes introducing selective denormalization, but the lead architect insists on maintaining full normalization for long-term flexibility. (5)
- A) Evaluate how the DBA can balance the two key physical design objectives, Integrity and Flexibility in this situation. Justify which design choices would best preserve structural stability without sacrificing system performance.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) During database deployment, the DBA uses a Composite Load Map to visualize how multiple processes access and update different data tables. (3)
Analyze how the Composite Load Map helps the DBA identify Referential Groups and define Unit of Work Constraints. Explain how these insights directly support implementation planning, particularly for managing backups, recovery points, and transaction consistency.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) In a product inventory system, each product passes through different stages during its life - such as *Created, Available, Out of Stock, and Discontinued*. (2)
Using the concept of Entity Life Cycle Analysis (ELCA), explain what rule determines when a product (entity) moves from one state to another. Provide one example of a valid state transition and one that would be invalid according to ELCA principles.

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 DS_3151:DATA ARCHITECTURE

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 DS_3151:DATA ARCHITECTURE

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	10(20%)
4	Analysis	5	15(30%)
5	Create	5	25(50%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)